

GOVIND GANDHI

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EDUCATION

PhD, Data Science

08/2022 - 05/2027 (Expected)

University of Virginia, Charlottesville, VA

3.86/4.0 GPA

School of Data Science

Coursework: Deep Learning, Natural Language Processing, Advanced Graph Neural Networks, Artificial Intelligence

BS in Physics, MS in Data Science

08/2017 - 05/2022

Indian Institute of Science Education and Research Pune, India

9.1/10.0 GPA

Best Interdisciplinary thesis awardee

Coursework: Nonlinear Dynamics, Probability and Statistics, Computational Physics, Economics & Public Policy

TECHNICAL SKILLS

Programming Languages: Python, C/C++, R, MATLAB, Perl, Julia, HTML, CSS, JavaScript, SQL

Libraries: PyTorch, Scikit-Learn, Hugging Face, CUDA, React, TensorFlow, VectorDB, Keras, NLTK

Certifications: NLP with Representation by DeepLearning.AI, Data Analyst Capstone project by IBM

PUBLICATIONS

- Gandhi, Govind, Ahn, Yong-Yeol, and Kojaku, Sadamori. "Self-supervised modularity maximization on graph embeddings for clustering." (under review)
- Akçakır, Gülşah, et al. "Exploring the interplay of individual traits and interaction dynamics in preschool social networks." arXiv preprint arXiv:2407.12728 (2024)
- Gandhi, Govind. "Symbolic Regression of Dynamic Network Models." arXiv preprint arXiv:2401.05369 (2023). (under review)
- Gandhi, Govind, and M. S. Santhanam. "Biased random walkers and extreme events on the edges of complex networks." Physical Review E 105.1 (2022): 014315. Jan 2022

PROJECTS AND EXPERIENCE

Principal Associate, Data Science - Intern: Capital One, McLean, VA

06/2025 - 08/2025

- Built the bank's first enterprise approach for **data-shift risk** in loss-forecasting—selecting **Bayesian Online Change-point Detection (BOCPD)** and implementing a production-ready Python workflow (ingest → detect → alert → evaluate).
- Proved it out** on synthetic stress suites and historical portfolios; **surfaced macro-regime shifts and portfolio-mix changes** earlier and with fewer false alarms than existing drift tests.
- Drove adoption:** aligned with Model Risk + business stakeholders, shipped runbook and training, demoed to senior execs; slated for rollout across **150+** developers/analysts/scientists enterprise-wide.

Visiting Research Fellow: University of Barcelona, Department of Physics, Barcelona, Spain 12/2024 – 06/2025

- Collaborated with Profs. Marián Boguñá and Mariangeles Serrano on applying hyperbolic graph embedding techniques (Mercator) to large-scale journal citation networks.
- Extracted and processed massive datasets from OpenAlex to model the flow and gatekeeping of ideas between journals in the scientific publication ecosystem.
- Analyzed author trajectories through hyperbolic embedding space to uncover patterns of interdisciplinary movement, community boundaries, and structural barriers to idea diffusion.
- Developed reproducible workflows for large-scale network embedding, combining computational physics, network science, and bibliometrics.

Consultant, AI Development Lead: Corto.ai, Los Angeles, CA

05/2024 - 12/2024

- Leading the development of advanced complex network-based representation learning models, leveraging social media data to identify patterns and insights for content recommendation and community detection.
- Spearheading integration of AI techniques with large-scale data processing pipelines, ensuring efficient handling of real-time social media streams.
- Collaborating with cross-functional teams to translate research innovations from the lab to the bench into scalable AI solutions, driving product development and enhancing user engagement strategies.

Assistant Researcher: Project Alt-Means, Luddy School of Informatics, Bloomington, IN 08/2023 - 08/2024

- Worked with Prof. Yong-Yeol Ahn and Prof. Sadamori Kojaku to develop a community detection algorithm using geometric deep learning - representation learning on graph neural network formalism.
- Formulated a novel clustering algorithm that outperforms K-Means without needing prior information of the number of clusters. **Code:** https://github.com/govgandhi/alt_means_sans_k
- Robust to parameter changes in graph embeddings and avoids bias towards homogeneous or large communities. Unveils effects of metric distances on embedding+clustering protocols.

Consultant: Workplace Behavioral Dynamics, Kelley School of Business, Bloomington, IN 06/2023 - 08/2023

- Analyzed the ENRON dataset using simple and complex-contagion network dynamics models to investigate the spread of unethical behaviour within the workplace.
- Analysed time series of interaction strength, sentiment analysis of email exchanges, and causal analysis of social and email interactions to identify communication patterns and ethical shifts, correlating them with key events and potential contagion triggers.
- Delivered actionable insights and a comprehensive report that informed academic and corporate governance strategies.

Assistant Researcher: Science Genome Project, Bloomington, IN 08/2022 - 08/2023

- Worked with Prof. Stasa Milojevic to quantify the extent of reliance of firms on basic sciences in their R&D efforts using deep learning methods.
- Mined, cleaned, and compiled semi-structured big data from Microsoft Academic Graph and Web of Science for analyzing patent-citation patterns.
- Consulted on a case provided by Grant Thornton International combining data analysis and business management to provide a market expansion strategy and change management protocol backed by network analysis of world trade networks using graph representation learning.

Risk & Financial Advisory Intern: Deloitte Corp. Finance Services, Bangalore, India 05/2022 - 08/2022

- Assisted in conducting risk assessments and preparing reports to help clients identify, manage, and mitigate risks.
- Participated in financial due diligence and regulatory compliance projects.
- Conducted data analysis and modeling to identify potential financial and operational risks.

Applied Scientist: Marc Bloch Center, CNRS, Berlin, Germany 07/2021 - 05/2022

- Worked with Prof. Camille Roth and Dr. Telmo Menezes to improve symbolic regression algorithms by developing novel evolutionary programming techniques.
- Optimized the algorithm to describe and predict the evolution of socio-semantic networks 120% faster.
- Converted the social network analysis into an open-source library. **Code:** <https://github.com/telmomenezes/synthetic>

Undergraduate Researcher: Complex Networks Group, IISER Pune, Pune, India 01/2020 – 07/2021

- **Project 1: Social Network Analysis on Gossip & Credibility Networks** to understand misinformation spreading.
- Worked with Prof. MS Santhanam to propose a novel social influence model involving the credibility of the sources to predict high-impact nodes.
- Tested the model on (online) Facebook friend networks and (offline) social networks in villages of Karnataka with conflicting pieces of information to find reinforcement of typical bipolarization behavior and positive feedback loops.
- **Project 2: Street traffic Optimization using Random Walkers**
- Developed a novel analytical framework with Prof. MS Santhanam, harnessing biased random walkers to study extreme events happening on the edges of complex networks and tested it on street networks. **Writing Sample:** [Extreme Events on Street Networks](#)

Undergraduate Researcher: Computational Neuroscience Lab, IISER Pune, Pune, India 05/2019 – 12/2019

- Under the supervision of Prof. Collins Assisi, modeled human grid cells as a Continuous Attractor Neural Network.
- Tested the Neural Network and quantified the role of Spike-Timing-Dependent Plasticity in enabling grid cells to spatially map out their environment.
- Culminated with public talk "Inhibitory Plasticity in Entorhinal Networks - How do we know the network has learned?" (IISER Pune, India - Nov 2019).

Public Policy Intern: IDFC Institute, Mumbai, India 05/2018 - 07/2018

- Designed and organized 25+ interviews in the slums of Dharavi and collaborated with a team of 15 to prepare the policy report: ["Decline of Rental Housing in India: The Case of Mumbai"](#) (Batra, Gandhi & Hingorani, 2018).

- Surveyed 250+, interviewed 60+, and worked with the stakeholders of IISERP's mental health while authoring "Blue & Lonesome" - G. Gandhi, K. Halder, I. Bezbarua (Kalpa, 2019).

LIFE IN ACADEMIA

Talks

- "Self-Supervised Modularity Maximization using graph embeddings", Oral presentation, NetSciX, IIT Indore, India, 2025
- "Does a consistent notion of distance improve clustering performance in graph embeddings?", Invited Talk, IISER Pune, India, 2024
- "Introduction to Network Science", Guest Lecture for INFO-I201, IU Bloomington, IN, USA
- "Introduction to Science of Science", IISER alumni talk series, IISER Pune, India, 2023
- "Creating and Using Centrality Measures to identify optimal GT office locations", Grant Thornton LLP, Chicago, IL, USA, 2022
- "Creating Artificial (network) scientists using Symbolic Regression", Master's thesis talk series, IISER Pune, India, 2022

Posters

- "Self-Supervised Modularity Maximization on Graph Embeddings", NetSci, Quebec city, Canada, 2024
- "Quantifying the growth of unique concepts in Science" - ISSI, IU Bloomington, IN, USA, 2023
- "Inhibitory Plasticity in Networks of the Entorhinal Cortex" - Annual Undergraduate Research Symposium - IISER Pune, India, 2019

Selected Honors

PhD Delegate, Graduate Data Science Council, University of Virginia, 2025-2026 AccelNet-MultiNet Fellow, program supported by NSF, NetSI & IUNI, 2024-2025 Graduate and Professional Student Government Doctoral Research Award, 2025 Jose Blakeley PhD Summer Research Award, 2024 INSPIRE Scholar, Govt. of India, 2017-2022 NTSE Scholar, Govt. of India, 2016-2021 Certificates of Merit in English, Mathematics and Computer Science, "Top 0.1% exam takers in the country", Govt. of India, 2017

Conferences/Workshops/Summer Schools

- Complexity72h 2024 - Madrid, Spain
- NetSci 2024 - Quebec City, Canada
- Lake Como Summer School on Complex Systems 2024 - Lake Como, Italy
- International Societ of Scientometrics and Infometrics (ISSI) 2023 - Bloomington, IN, USA

Teaching

- Teaching Assistant, IU Bloomington: INFO-I 201 Mathematical Foundations of Informatics